or even the sacrificing of one's life for abstract causes could all be dismissed (as many neoclassical economists dismiss them) if they were isolated events. But obviously they are not and they must be taken into account if we are to advance our understanding of human behavior. If our understanding of motivation is very incomplete, we can still take an important forward step by taking explicit account of the way institutions alter the price paid for one's convictions and hence play a critical role in the extent to which nonwealth-maximizing motivations influence choices. We will take such account in succeeding chapters. But first we must examine in detail what it is about the environment that is so complex.

4

A transaction cost theory of exchange

My theory of institutions is constructed from a theory of human behavior combined with a theory of the costs of transacting. When we combine them we can understand why institutions exist and what role they play in the functioning of societies. If we add a theory of production we can then analyze the role of institutions in the performance of economies.

The costliness of information is the key to the costs of transacting, which consist of the costs of measuring the valuable attributes of what is being exchanged and the costs of protecting rights and policing and enforcing agreements. These measurement and enforcement costs are the sources of social, political, and economic institutions. The rest of this chapter concentrates on economic exchange; in Chapter 6 I will build a model of political exchange from the same building blocks.

The costliness of economic exchange distinguishes the transaction costs approach from the traditional theory economists have inherited from Adam Smith. For 200 years the gains from trade made possible by increasing specialization and division of labor have been the cornerstone of economic theory. Specialization could be realized by increasing the size of markets, and as the world's economy grew and division of labor became ever more specific, the number of exchanges involved in the performance of economies expanded. But the long line of economists who built this approach into an elegant body of economic theory did so without regard to the costliness of this exchange process. An exchange process involving transaction costs suggests significant modifications in economic theory and very different implications for economic performance.[1]

[1]The transaction cost approach is consistent only in its agreement on the importance of transaction costs; it is far from unified in other respects. The approach developed here might most appropriately be characterized as the University of Washington approach, originated by Steven Cheung (1974, 1983) and elaborated, modified, and developed at the University of Washington, most notably by Yoram Barzel (1982, 1989) but also by Keith Leffler (with Klein, 1981), Masanori Hashimoto (1979), and

Wallis and North (1986), measuring the size of transaction costs that go through the market (such as costs associated with banking, insurance, finance, wholesale, and retail trade; or, in terms of occupations, with lawyers, accountants, etc.) in the U.S. economy found that more than 45 percent of national income was devoted to transacting and, moreover, that this percentage had increased from approximately 25 percent a century earlier. Thus the resources of the economy consumed in transacting are of considerable magnitude and growing. Because transaction costs are a part of the costs of production, we need to restate the traditional production relationship as follows. The total costs of production consist of the resource inputs of land, labor, and capital involved both in transforming the physical attributes of a good (size, weight, color, location, chemical composition, and so forth) and in transacting – defining, protecting, and enforcing the property rights to goods (the right to use, the right to derive income from the use of, the right to exclude, and the right to exchange).

Once we recognize that the costs of production are the sum of transformation and transaction costs, we require a new analytical framework of microeconomic theory.[2] However, our concern in this study is a theory of institutions, and although that focus inevitably overlaps with some fundamental issues in microeconomic theory, to explore systematically the implications for the latter theory would take us in another direction. Our initial question, however – why is it costly to transact? – is common both to the restructuring of microtheory and to a theory of institutions.

I

As we saw in Chapter 2, in "The Problem of Social Cost" (1960) Ronald Coase made clear that only in the absence of transaction costs did the neoclassical paradigm yield the implied allocative results; with positive transaction costs, resource allocations are altered by property rights structures. Neither Coase nor many of the subsequent studies of transaction costs have attempted to define precisely what it is about transacting that is so costly, but that issue is central to the issues of this study and I now turn to it. I begin by exploring the costliness of measurement (holding enforcement costs constant) and then in Section III examine the costs of enforcement.

We get utility from the diverse attributes of a good or service or, in the case of the performance of an agent, from the multitude of separate

Douglass North (1981, 1984). Other approaches, notably that of Oliver Williamson, will be contrasted with the approach developed here.

[2]For the beginning of such a theory, see Barzel (1989).

activities that constitute performance.[3] This means, in common sense terms, that when I consume orange juice, I get utility from the quantity of juice I drink, the amount of vitamin C it contains, and its flavor, even though the exchange itself consisted simply of paying $2.00 for fourteen oranges. Similarly, when I buy an automobile, I get a particular color, acceleration, style, interior design, leg room, gasoline mileage – all valued attributes, even though it is only an automobile I buy. When I buy the services of doctors, their skill and bedside manner and the time spent waiting in their offices are part of the purchase. When as chairman of an economies department I hire assistant professors, not only the quantity and quality (however measured) of their teaching and research output (again, however measured), but a multitude of other aspects of their performance are also hired: whether they prepare for and meet classes on time, provide external benefits to colleagues, cooperate in department affairs, do not abuse their positions vis-à-vis students, or call friends in Hong Kong at departmental expense. The value of an exchange to the parties, then, is the value of the different attributes lumped into the good or service. It takes resources to measure these attributes and additional resources to define and to measure rights that are transferred.

The transfers that occur with an exchange entail costs that result from both parties attempting to determine what the valued attributes of these assets are – attributes that, because of prohibitive measurement costs, have remained poorly delineated. Thus, as a buyer of oranges I attempt to purchase an amount of juice, an amount of vitamin C, and the flavor of oranges, even though what I purchased was simply fourteen oranges for $2.00. Similarly, when as a potential buyer I look at an automobile, I attempt to ascertain whether it has the attributes important to me in a car. The same holds for the purchase of a doctor's services, about which I try to ascertain information on skill, bedside manner, and office waiting time.

From the particulars in the foregoing illustrations we can generalize as follows: commodities, services, and the performance of agents have numerous attributes and their levels vary from one specimen or agent to another. The measurement of these levels is too costly to be comprehensive or fully accurate. The information costs in ascertaining the level of individual attributes of each unit exchanged underlie the costliness of this aspect of transacting. Even if all exchanging individuals had the same objective function (for example, jointly maximizing the wealth of a firm that employed them), there would still be the transaction costs involved in acquiring the necessary information about the levels of attributes of each exchange unit, the location of buyers (sellers), and so forth. But, in fact

[3]For this extension of consumer theory see Lancaster (1966), Becker (1965), Cheung (1974), and Barzel (1982), among others.

there are asymmetries of information among the players, and these and the underlying behavioral function of individuals in combination produce radical implications for economic theory and for the study of institutions.

I take up the issue of asymmetry first. In the foregoing illustrations, the seller of oranges knew much more about the valuable attributes of the oranges than the buyer, the used car dealer knew much more about the valued attributes of the car than the buyer (Akerlof, 1970), and the doctor knew much more about the quality of services and skill than the patient. Likewise, prospective assistant professors know much more about their work habits than does the department chairman or, to take another example, the purchaser of life insurance from an insurance company knows much more about his or her health than the insurer does.

Not only does one party know more about some valued attribute than the other party, he or she may stand to gain by concealing that information. According to a strictly wealth-maximizing behavioral assumption, a party to exchange will cheat, steal, or lie when the payoff to such activity exceeds the value of the alternative opportunities available to the party. Indeed, this assumption was the basis of Akerlof's famous article on lemons (1970), of the dilemmas posed by adverse selection in the purchase of life insurance, of problems of moral hazard (Holmstrom, 1979), and of a multitude of other issues that have emerged in the literature over the last dozen years in what is called the New Industrial Organization literature. Although it is sometimes in the interests of the exchanging parties to conceal certain kinds of information, at other times it is in their interests to reveal information. With this background we can develop some generalizations about the measurement aspects of a transaction cost model of exchange.

II

Consider first the standard neoclassical Walrasian model. In this general equilibrium model, commodities are identical, the market is concentrated at a single point in space, and the exchange is instantaneous. Moreover, individuals are fully informed about the exchange commodity and the terms of trade are known to both parties. As a result, no effort is required to effect exchange other than to dispense with the appropriate amount of cash. Prices, then, become a sufficient allocative device to achieve highest value uses.

To the Walrasian model, which includes the maximizing behavior of individuals, the gains that result from specialization, and the division of labor that produces exchange, I now add costs of information. As noted above, these include the costs of measuring the valued attributes of goods

and services and the varying characteristics of the performance of agents. The net gains from exchange are the gross gains, which are the standard gains in neoclassical theory and in the international trade model, minus the costs of measuring and policing the agreement and minus the losses that result because monitoring is not perfect. On a common sense level, it is easy to see that we devote substantial resources and efforts to the measurement, enforcement, and the policing of agreements. Warranties, guarantees, trademarks, the resources devoted to sorting and grading, time and motion studies, the bonding of agents, arbitration, mediation, and of course the entire system of judicial process all reflect the ubiquity of measurement and enforcement.

Because it is costly to measure the valued attributes fully, the opportunity for wealth capture by devoting resources to acquiring more information is ever present. For example, the seller of a commodity such as fruits and vegetables may find it too costly to sort and grade them precisely. On the other hand, a buyer may find that it is worthwhile to devote time to picking and choosing among the fruit and vegetables available. In this case the seller has put into the public domain the variability of attributes that can in part be captured by the buyer devoting time and effort to sorting them out. The same can be said for the purchaser of a used automobile or the purchaser of medical services of doctors. Because of the enormous variety in the characteristics and the costliness of measuring attributes of goods and services and the performance of agents, the ideal ownership rights, with respect to these assets and resources, may take a variety of forms. In some cases, the ideal form is that the rights be divided among the parties. The buyer of a durable good, for example, may own some rights; others remain with the manufacturer in the form of guarantees of performance.

As a generalization, the more easily others can affect the income flow from someone's assets without bearing the full costs of their action, the lower is the value of that asset. As a result, the maximization of an asset's value involves the ownership structure in which those parties who can influence the variability of particular attributes become residual claimants over those attributes. In effect they are then responsible for their actions and have an incentive to maximize the potential gains from exchange. The rights to an asset generating a flow of services are usually easy to assure when the flow can be easily measured, because it is easy to impose a charge commensurate with a level of service. Therefore, when a flow is known and constant, it is easy to assure rights. If the flow varies but is predictable, rights are still easy to assure. When the flow of income from an asset can be affected by the exchange parties, assigning ownership becomes more problematic. When the income stream is variable and not fully predictable, it is costly to determine whether the flow is what it

should be in that particular case. In such an instance, both parties will try to capture some part of the contestable income stream.

III

So far the emphasis of the analysis has been on measurement. It is, however, measurement plus the costliness of enforcement that together determine the costs of transacting. If we return to the Walrasian model described above, we assume that there are no costs associated with enforcement of agreements. Indeed, as long as we maintain the fiction of a unidimensional good transacted instantaneously, the problems of policing and enforcement are trivial. But when we add the costs of acquiring information and, specifically, of measuring, the problems become major ones. It is because we do not know the attributes of a good or service or all the characteristics of the performance of agents and because we have to devote costly resources to try to measure and monitor them that enforcement issues do arise.

One issue is that of policing agents. The most extreme example concerns the relationship between a master and slave. There is, in fact, an implicit contract between the two; to get maximum effort from the slave, the owner must devote resources to monitoring and metering a slave's output and critically applying rewards and punishments based on performance. Because there are increasing marginal costs to measuring and policing performance, the master will stop short of perfect policing and will engage instead in policing until the marginal costs equal the additional marginal benefits from such activity. The result is that slaves acquire certain property rights in their own labor. That is, owners are able to enhance the value of their property by granting slaves some rights in exchange for services the owners value more. Hence slaves became owners too. Indeed it is only this ownership that made it possible for slaves to purchase their own freedom, as was frequently done in classical times and even occasionally in the antebellum South.[4]

Although the slave example is an extreme form, the agency issue is ubiquitous in hierarchical organizations. The problems of monitoring and metering the various attributes that constitute performance of agents mean that, in contrast to the standard neoclassical frictionless model of workers being paid the value of their marginal product, they are paid this cost minus the resource costs of monitoring and policing.[5] In the above illustration I implicitly introduced property rights when I referred to the

[4]See Barzel (1977) for a detailed elaboration of this argument.

[5]Jensen and Meckling in a well-known essay (1976) have elaborated on the agency costs involved in monitoring, policing, and the shirking of agents.

concept of a master owning a slave; and in all discussions of principal/agents and the monitoring problem, we assume that the principal has the power of disciplining the agent and therefore of enforcing agreements. Likewise, the agent can monitor the principal and enforce his or her end of the agreement.

Enforcement can come from second-party retaliation. It also can result from internally enforced codes of conduct or by societal sanctions or a coercive third party (the state).

But one cannot take enforcement for granted. It is (and always has been) the critical obstacle to increasing specialization and division of labor. Enforcement poses no problem when it is in the interests of the other party to live up to agreements. But without institutional constraints, self-interested behavior will foreclose complex exchange, because of the uncertainty that the other party will find it in his or her interest to live up to the agreement. The transaction cost will reflect the uncertainty by including a risk premium, the magnitude of which will turn on the likelihood of defection by the other party and the consequent cost to the first party. Throughout history the size of this premium has largely foreclosed complex exchange and therefore limited the possibilities of economic growth.

IV

We are now ready to explore the relationship among the behavioral assumptions developed in Chapter 3, the characteristics of transacting as developed in the previous sections of this chapter, and the institutional structure of a society.

Property rights are the rights individuals appropriate over their own labor and the goods and services they possess. Appropriation is a function of legal rules, organizational forms, enforcement, and norms of behavior – that is, the institutional framework. Because with any property rights structure transaction costs are positive, rights are never perfectly specified and enforced; some valued attributes are in the public domain and it pays individuals to devote resources to their capture. Because the costs of transacting have changed radically throughout history and vary equally radically in different contemporary economies, the mix between the formal protection of rights and individual attempts to capture rights or devote resources to individual protection of their own rights varies enormously. We have only to compare property rights in Beirut in the 1980s with those of a modern small-town U.S. community to cover the spectrum. In the former, most valuable rights are in the public domain, to be seized by those with the violence potential to be successful; in the latter the legal structure defines and enforces a large share of rights and those

valuable rights in the public domain tend to be allocated by traditional norms of behavior. The difference between these two is a function of the institutional structure in each.

Institutions provide the structure for exchange that (together with the technology employed) determines the cost of transacting and the cost of transformation. How well institutions solve the problems of coordination and production is determined by the motivation of the players (their utility function), the complexity of the environment, and the ability of the players to decipher and order the environment (measurement and enforcement).

The institutions necessary to accomplish economic exchange vary in their complexity, from those that solve simple exchange problems to ones that extend across space and time and numerous individuals. The degree of complexity in economic exchange is a function of the level of contracts necessary to undertake exchange in economies of various degrees of specialization. Nonspecialization is a form of insurance when the costs and uncertainties of transacting are high. The greater the specialization and the number and variability of valuable attributes, the more weight must be put on reliable institutions that allow individuals to engage in complex contracting with a minimum of uncertainty about whether the terms of the contract can be realized. Exchange in modern economies consisting of many variable attributes extending over long periods of time necessitates institutional reliability, which has only gradually emerged in Western economies. There is nothing automatic about the evolution of cooperation from simple forms of contracting and exchange to the complex forms that have characterized the successful economies of modern times.

Institutions structure economic exchange in an enormous variety of forms that do, however, fall into general types that are consistent with the transactions cost model of exchange. The kind of exchange that has characterized most of economic history has been personalized exchange involving small-scale production and local trade. Repeat dealing, cultural homogeneity (that is a common set of values), and a lack of third-party enforcement (and indeed little need for it) have been typical conditions. Under them transactions costs are low, but because specialization and division of labor is rudimentary, transformation costs are high. The economies or collections of trading partners in this kind of exchange tend to be small.

As the size and scope of exchange have increased, the parties have attempted to clientize or personalize exchange. But the greater the variety and numbers of exchange, the more complex the kinds of agreements that have to be made, and so the more difficult it is to do. Therefore a second general pattern of exchange has evolved, that is impersonal exchange, in which the parties are constrained by kinship ties, bonding, exchanging

hostages, or merchant codes of conduct. Frequently the exchange is set within a context of elaborate rituals and religious precepts to constrain the participants. The early development of long-distance and cross-cultural trade and the fairs of medieval Europe were built on such institutional constructs. They permitted a widening of the market and the realization of the gains from more complex production and exchange, extending beyond the bounds of a small geographic entity. In early modern Europe, these institutions led to an increasing role of the state in protecting merchants and to the adoption of merchant codes as the revenue potential of such fiscal activities increased. However, in this environment the role of the state was at best ambiguous, because the state was as often an increasing source of insecurity and higher transaction costs as it was protector and enforcer of property rights.

The third form of exchange is impersonal exchange with third-party enforcement. It has been the critical underpinning of successful modern economies involved in the complex contracting necessary for modern economic growth. Third-party enforcement is never ideal, never perfect, and the parties to exchange still devote immense resources to attempting to clientize exchange relationships. But neither self-enforcement by parties nor trust can be completely successful. It is not that ideology or norms do not matter; they do and immense resources are devoted to attempting to promulgate codes of conduct. Equally, however, the returns on opportunism, cheating, and shirking rise in complex societies. A coercive third party is essential. One cannot have the productivity of a modern high income society with political anarchy. Indeed, effective third-party enforcement is best realized by creating a set of rules that then make a variety of informal constraints effective. Nevertheless, the problems of achieving third-party enforcement of agreements via an effective judicial system that applies, however imperfectly, the rules are only very imperfectly understood and are a major dilemma in the study of institutional evolution.

Thus, it should be readily apparent that to develop a model of institutions, we must explore in depth the structural characteristics of informal constraints, formal rules, and enforcement and the way in which they evolve. Then we shall be in a position to put them together to look at the overall institutional makeup of political/economic orders.

5

Informal constraints

In all societies from the most primitive to the most advanced, people impose constraints upon themselves to give a structure to their relations with others. Under conditions of limited information and limited computational ability, constraints reduce the costs of human interaction as compared to a world of no institutions. However, it is much easier to describe and be precise about the formal rules that societies devise than to describe and be precise about the informal ways by which human beings have structured human interaction. But although they defy, for the most part, neat specification and it is extremely difficult to develop unambiguous tests of their significance, they are important.

In the modern Western world, we think of life and the economy as being ordered by formal laws and property rights. Yet formal rules, in even the most developed economy, make up a small (although very important) part of the sum of constraints that shape choices; a moment's reflection should suggest to us the pervasiveness of informal constraints. In our daily interaction with others, whether within the family, in external social relations, or in business activities, the governing structure is overwhelmingly defined by codes of conduct, norms of behavior, and conventions. Underlying these informal constraints are formal rules, but these are seldom the obvious and immediate source of choice in daily interactions.

That the informal constraints are important in themselves (and not simply as appendages to formal rules) can be observed from the evidence that the same formal rules and/or constitutions imposed on different societies produce different outcomes. And discontinuous institutional change, such as revolution or military conquest and subjugation, certainly produces new outcomes. But what is most striking (although seldom observed, particularly by advocates of revolution) is the persistence of so many aspects of a society in spite of a total change in the rules. Japanese culture survived the U.S. occupation after World War II; the post-revolutionary U.S. society remained much as it had been in colonial

times; Jews, Kurds, and endless other groups have persisted through centuries despite endless changes in their formal status. Even the Russian Revolution, perhaps the most complete formal transformation of a society we know, cannot be completely understood without exploring the survival and persistence of many informal constraints.

Where do informal constraints come from? They come from socially transmitted information and are a part of the heritage that we call culture. The way the mind processes information depends "upon the brain's ability to learn by being programmed with one or more elaborately structured natural languages that can code for perceptual, attitudinal and moral (behavioral) as well as factual information" (Johansson, 1988, p. 176). Culture can be defined as the "transmission from one generation to the next, via teaching and imitation, of knowledge, values, and other factors that influence behavior" (Boyd and Richerson, 1985, p. 2). Culture provides a language-based conceptual framework for encoding and interpreting the information that the senses are presenting to the brain.

Essentially the argument being made here is an extension of the argument of Chapter 3 that processing information is the key to understanding a more complex behavioral pattern than is derived from the expected utility model. But the emphasis in that chapter was on the incompleteness of the information and the consequent need for institutions to structure human interrelations. In this chapter the emphasis is on the way that the cultural filter provides continuity so that the informal solution to exchange problems in the past carries over into the present and makes those informal constraints important sources of continuity in long-run societal change.

I

I begin by examining human interaction when there are no formal rules. How is order preserved in stateless societies? The anthropological literature is extensive, and although many of the findings are still controversial, it makes important reading not only for the study of historical work and for an analysis of order in primitive societies, but also for its implications for a modern understanding of informal constraints. Robert Bates (1987), building his analysis on Evans-Pritchard's classic study of the Nuer, states the issue of such constraints as follows:

> The puzzle, from Evans-Pritchard's point of view, was that, despite the potential for theft and disorder, the Nuer in fact tended to live in relative harmony. Insofar as the Nuer raided cattle, they tended to raid the cattle of others; raids within the tribe were relatively rare. Somehow the Nuer appear to have avoided the potentially harmful effects arising from the pursuit of self-interest. And they appear to

have done so even while lacking those formal institutions so common in Western societies which specialize in preserving the peace and forestalling violence: the courts, the police, and so on. (Bates, 1987, p. 8)

Bates then describes the deterring effects that both compensation among the tribe and the threat of feud posed for preserving order. He shows how this cooperative solution makes sense in game theoretic terms. A one-shot prisoner's dilemma problem, where it would appear that the players must arrive at a violent solution with the result that each family is worse off, is avoided. Instead an iterated game is played, and with the threat of feud it is in the interests of the parties to preserve order and hence not to pursue interfamily cattle raiding. The critical point here is that it is the members of the family themselves who prevent other family members from engaging in raiding, because a feud, once started, would be harmful to all members.

The extensive literature that anthropologists have produced on primitive societies makes clear that exchange in tribal societies is not simple. In the absence of the state and formal rules, a dense social network leads to the development of informal structures with substantial stability. No one has described this situation better than Elizabeth Colson (1974).

> Whether we call them customs, laws, usages, or normative rules seems of little importance. What is important is that communities such as the Tonga do not leave their members free to go their own way and explore every possible avenue of behavior. They operate with a set of rules or standards which define appropriate action under a variety of circumstances. The rules, by and large, operate to eliminate conflict of interests by defining what it is people can expect from certain of their fellows. This has the healthy effect of limiting demands and allowing the public to judge performance. . . .
>
> At another level, however, they would see conflict as endemic to social life because people who live in close juxtaposition use the same space and want support and attention from the same individuals. Rules, even though they may at times produce conflict, reduce the chances for conflict because they reduce the total amount of ambiguity for those concerned by defining specific rather than universalistic claims and obligations. It becomes possible to order one's life with a set of priorities regarded as legitimate. . . . Among the Tonga I have had to learn that I should not give just because I feel like giving as this is an insult to all who do not receive. Rules do not solve all problems; they only simplify life.
>
> They also give a framework for organizing activities. Standards and some means of applying sanctions are necessary complements to the rules if a system of social control is to operate within a community. Among people such as the Tonga, onlookers apply the standards of performance in particular roles in making an overall judgement [sic] about the total person; this in turn allows them to predict future behavior. Judgment is an ongoing process through which consensus is finally reached. (Colson, 1974, pp. 51–3)

Several implications are clear from this review of work by Colson and other anthropologists. Order in the societies they describe is the result of a dense social network where people have an intimate understanding of each other and the threat of violence is a continuous force for preserving order because of its implications for other members of society. Deviant behavior cannot be tolerated in such a situation, because it is a fundamental threat to the stability and insurance features of the tribal group.

Richard Posner's model of primitive society (1980), which generates an explanation of many institutional features of such societies, is similar to the one I develop here (although mine has none of the maximizing social wealth or efficiency implications that are explicit in Posner's work). In Posner's model, high information costs, the absence of effective government, limited numbers of goods and limited trade, limited food preservation, and negligible gains from innovation produce a set of common characteristics:

> Weak government, ascription of rights and duties on the basis of family membership, gift-giving as a fundamental mode of exchange, strict liability for injuries, emphasis on generosity and honor as high ethical norms, collective guilt – these and other features of social organization recur with such frequency in accounts of primitive and archaic societies as to suggest that a simple model of primitive society, which abstracts from many of the particular features of specific societies, may nonetheless explain much of the structure of primitive social institutions. (Posner, 1980, p. 8)

Posner's essay emphasizes the importance of kinship ties as the central insurance, protection, and law enforcement mechanisms of primitive societies. Bates' study of Kenya (1989) equally focuses on the changing pattern of kinship ties in the context of political/economic conditions as the key to understanding the evolving institutional constraints of a society in rapid transition from a tribal society to a market economy.

II

Informal constraints are pervasive features of modern economies as well. In order to dispel the assertion by law and economics scholars of the centrality of legal doctrine, Robert Ellickson did a field study of the way in which rural residents of Shasta County, California, resolved disputes arising over trespass damage done by stray livestock.[1] He found that the residents almost never resorted to legal redress, but instead relied on an elaborate structure of informal constraints to resolve disputes. In a subsequent article (1987) and forthcoming book, Ellickson provides a great

[1] Appropriately titled "Of Coase and Cattle: Dispute Resolution Among Neighbors in Shasta County" (1986).

deal of additional empirical evidence of the pervasiveness of informal constraints.

Even the most casual introspection suggests the pervasiveness of informal constraints. Arising to coordinate repeated human interaction, they are (1) extensions, elaborations, and modifications of formal rules, (2) socially sanctioned norms of behavior, and (3) internally enforced standards of conduct. I elaborate on each of these aspects of informal constraints.

1. In a study of the institutional foundations of committee power Shepsle and Weingast (1987) demonstrate that the power of congressional committees that is not explained by the formal rules is a result of a set of informal unwritten constraints that have evolved in the context of repeated interaction (exchange) among the players. These constraints evolved from the formal rules to deal with specific problems of exchange and became established as recognized institutional constraints even though they were never made a part of the formal rules. Committee chairs and committees consequently have an influence over legislative choices that could not be derived from the formal structure.

2. Robert Axelrod (1986) provides a vivid illustration of a socially sanctioned norm of behavior. The night before he was to engage in a duel with Aaron Burr, Alexander Hamilton sat down and wrote out all the reasons why he should not accept this challenge; a crucial one, of course, was that he was likely to get killed. Yet, in spite of the overwhelming rational bases for not dueling, he felt that his effectiveness in the public arena would be significantly diminished by such a decision because dueling was the accepted way to settle disputes among gentlemen. Social norms dictated the choice, not formal rules.

3. Both of the first two types of informal constraints can be modeled in the context of wealth-maximizing models and therefore lend themselves to treatment in neoclassical (and game theory) frameworks. But internally enforced codes of conduct only have meaning in terms of informal constraints, altering choices when the individual gives up wealth or income for some other value in his or her utility function. Numerous essays explore voting behavior by legislators and conclude that one cannot explain legislative voting behavior by an interest group model (in which the legislator faithfully mirrors the interests of his or her constituents), but must take into account the subjective, personal preferences of the legislator (Kalt and Zupan, 1984). This literature is controversial because of the statistical problems in getting unambiguous answers, but there is abundant qualitative and quantitative evidence that the lower the price of ideas, ideologies, and convictions, the more they matter and affect choices (for empirical support see Nelson and Silberberg, 1987).

III

How do we explain the emergence and persistence of informal constraints? A pervasive but relatively simple to explain form of such constraints is conventions that solve coordination problems: "These are rules that have never been consciously designed and that it is in everyone's interest to keep" (Sugden, 1986, p. 54). The usual illustration of such a convention is *rules* of the road. The important characteristic of conventions is that, given the costs of exchange (Chapter 4), both parties have a stake in minimizing the costliness of measurement and the exchanges are self-enforcing. In terms of the total resources that go into transacting in an economy, conventions that solve coordination probably account for a larger proportion of the costs of transacting than the other informal constraints described later in this chapter (although in many instances the transaction costs in fact reflect a combination of sources of informal constraints).

Informal constraints that arise in the context of exchange but are not self-enforcing are more complex because they necessarily entail features that make the exchange viable by reducing measurement and enforcement costs. In the absence of constraints, asymmetric information and the consequent distribution of the gains will lead to devoting excessive resources to measurement or indeed can lead to exchange not taking place at all because the exchange is unenforceable. Informal constraints can take the form of agreed upon lower cost forms of measurement (standardized weights and measures, for example) and make second- and third-party enforcement effective by specific sanctioning devices or information networks that acquaint third parties with exchange performance (credit ratings, better business bureaus, etc.). Such organizations and instruments that make norms of cooperative behavior (informal constraints) effective are not only a major part of the story of more complex exchange through history, but are strikingly paralleled by the game theoretic models that produce cooperative outcomes through features that alter discount rates and increase information. The growth of more complex forms of exchange in later medieval and early modern Europe was made possible by a variety of informal institutions such as the early law merchant's publicized codes of merchant conduct. Prices current and the development of auditing and accounting techniques lowered critical (i.e., measurable) information and enforcement costs. These can be modeled in a game theoretic framework by raising the gains from cooperative action or raising the costs of defecting (see Milgrom, North, and Weingast, 1990).

Much more difficult to deal with in theoretical terms than wealth

maximizing informal constraints are internally enforced codes of conduct that modify behavior. It is difficult because one must devise a model that predicts choices in the context of the trade-off between wealth and other values. But strong religious beliefs or commitment to communism, for example, provide us with historical accounts of the sacrifices individuals have made for beliefs. As noted earlier, experimental economics provides evidence that individuals do not always free-ride and a study by Frank (1988) provides both a large body of evidence and a model of such behavior.

The literature cited above and the earlier chapter of this book dealing with human behavior make clear that motivation is more complicated than the simple expected utility model. Chapter 3 also emphasized that under certain conditions traits like honesty, integrity, and living up to a reputation pay off in strictly wealth-maximizing terms. Still unexplained is a very large residual. We simply do not have any convincing theory of the sociology of knowledge that accounts for the effectiveness (or ineffectiveness) of organized ideologies or accounts for choices made when the payoffs to honesty, integrity, working hard, or voting are negative.

Two partial explanations are Howard Margolis's (1982) dual utility model (mentioned in Chapter 2) and Robert Sugden's argument about conventions acquiring moral force. Margolis's argument is that individuals possess not one but two utility functions: *S* preferences are governed by the usual self-interest preference function, whereas *G* preferences are purely social (group interested). Margolis attempts to give empirical content to the argument by developing a model with assigned weights given self-interest preferences versus weights given group-interest preferences and by exploring the conditions under which the weights change. Sugden (1986) maintains that a convention acquires moral force when almost everyone in the community follows it, and it is in the interests of each individual that people with whom he or she deals follow the rule providing that the individual does too. What evolves according to Sugden is a "morality of cooperation" (Sugden, 1986, p. 173).

IV

It is time to pull together and summarize the argument of this chapter. The way by which the mind processes information not only is the basis for the existence of institutions, but is a key to understanding the way informal constraints play an important role in the makeup of the choice set both in the short-run and in the long-run evolution of societies.

In the short run, culture defines the way individuals process and utilize information and hence may affect the way informal constraints get specified. Conventions are culture specific, as indeed are norms. However

norms pose some still unexplained problems. What is it that makes norms evolve or disappear – for example, dueling as a solution to gentlemanly differences?

Even if we do not possess a good explanation for social norms, we can model wealth-maximizing norms in a game theoretic context. That is, we can explore and test, empirically, what sorts of informal constraints are most likely to produce cooperative behavior or how incremental changes in such informal constraints will alter the game to increase (or decrease) cooperative outcomes. This approach may increase our understanding of the development of more complex forms of exchange, such as the early evolution of financial markets.[2]

A transaction cost framework equally offers promise of exploring informal constraints. Although the informal institutional constraints are not directly observable, the contracts that are written, and sometimes the actual costs of transacting, provide us with indirect evidence of changes in informal constraints. The striking decline in interest rates in the Dutch capital market in the seventeenth century and in the English capital market in the early eighteenth century provides evidence of the increasing security of property rights as a consequence of the effective interaction of a variety of both formal and informal institutional constraints. For example, the enforcement of contracts that evolved from merchant codes of behavior included ostracism of those who violated agreements and the eventual encoding of customary practices into the formal law.[3]

The importance of self-imposed codes of behavior in constraining maximizing behavior in many contexts also is evident. Our understanding of the source of such behavior is deficient, but we can frequently measure its significance in choices by empirically examining marginal changes in the cost of expressing convictions. Such analysis opens the door to explaining the power of subjective perceptions in affecting choices. If the demand function is negatively sloped (i.e., the lower the cost of expressing one's convictions the more important will the convictions be as a determinant of choice) and formal institutions make it possible for individuals to express preferences at little cost to themselves, then indeed the subjective preferences that individuals hold play a big part in determining choices. Voting, hierarchies that produce slack in the principal/agent relationship in legislatures, and lifetime tenure for judges are formal institutional constraints that lower the cost of acting on one's convictions.

It is simply impossible to make sense out of history (or contemporary

[2]For an interesting game theoretic application, see John Veitch, "Repudiations and Confiscations by the Medieval State" (1986).

[3]See Douglass C. North, "Institutions, Transaction Costs, and the Rise of Merchant Empires," in J. Tracy, editor, *The Political Economy of Merchant Empires,* Cambridge University Press (forthcoming).

economies) without recognizing the central role that subjective preferences play in the context of formal institutional constraints that enable us to express our convictions at zero or very little cost. Ideas, organized ideologies, and even religious zealotry play major roles in shaping societies and economies.

Nineteenth-century U.S. economic history, briefly described in Chapter 1, is full of illustrations. Whether we trace the history and consequences of the abolitionist movement, or examine the reasoning of Supreme Court justices that explicitly undergirded the decisions they handed down, or explore the organization, policies, and legislative enactments of the Greenback, Granger, and Populist movements of the U.S. farmer, they all only make sense in the context of subjective perceptions of the actors in the context of formal institutional structures that altered the price individuals paid for their convictions and hence enabled their choices to become effective.

In the first case the religious zealotry of the abolitionist groups that activated them to organize politically, together with the Northern electorate's growing conviction of the immorality of slavery and the 1860 election, led to the Civil War and the elimination of slavery (Fogel, 1989). In the second case the lifetime tenure specified for justices shielded them from interest group pressures and permitted – encouraged – them to vote their convictions. Their convictions were derived from their subjective construction of the issues. From the Marshall Court (1801 to 1835) to the Rehnquist Court, the justices have interpreted and reinterpreted essentially the same set of rules. The Court reverses itself 180 degrees over time because the judges' subjective modeling of the issues changes. The third case reflects farmers' persistent beliefs that they were being wronged by monetary policies, railroads, grain elevators, bankers, and others. They acted on these convictions by forming organizations with the objective of enacting *corrective* legislation first in state legislatures, then through the Populist party and the Democratic party in the U.S. Congress.

What determines how much people will pay to express and act on their convictions? We seldom know much about the elasticity of the function or shifts in the function, but we do have abundant evidence that the function is negatively sloped and that the price incurred for acting on one's convictions is frequently very low (and hence convictions are significant) in many institutional settings.

The long-run implication of the cultural processing of information that underlies informal constraints is that it plays an important role in the incremental way by which institutions evolve and hence is a source of path dependence. We still are a long way from having any neat models of cultural evolution (although see Cavalli-Sforza and Feldman, 1981, and Boyd and Richerson, 1985, for some interesting attempts), but we do

know that cultural traits have tenacious survival ability and that most cultural changes are incremental.

Equally important is the fact that the informal constraints that are culturally derived will not change immediately in reaction to changes in the formal rules. As a result the tension between altered formal rules and the persisting informal constraints produces outcomes that have important implications for the way economies change, which is the subject of Part II.